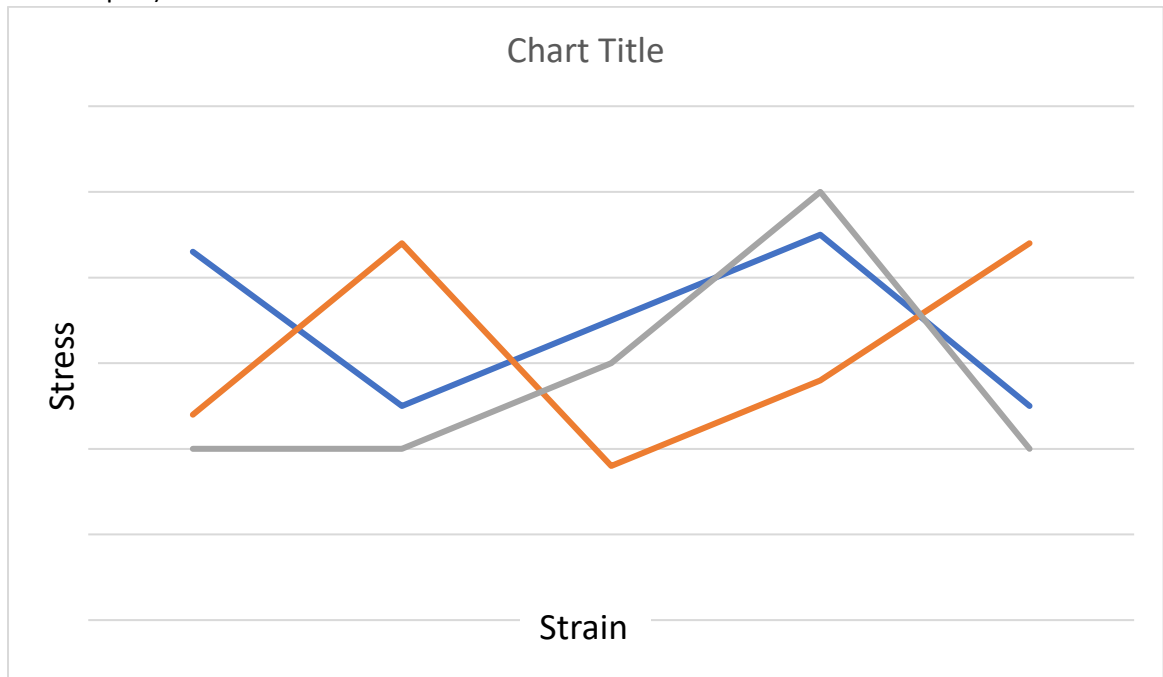


Tensile Test Report Template

Specimen 1: Metal

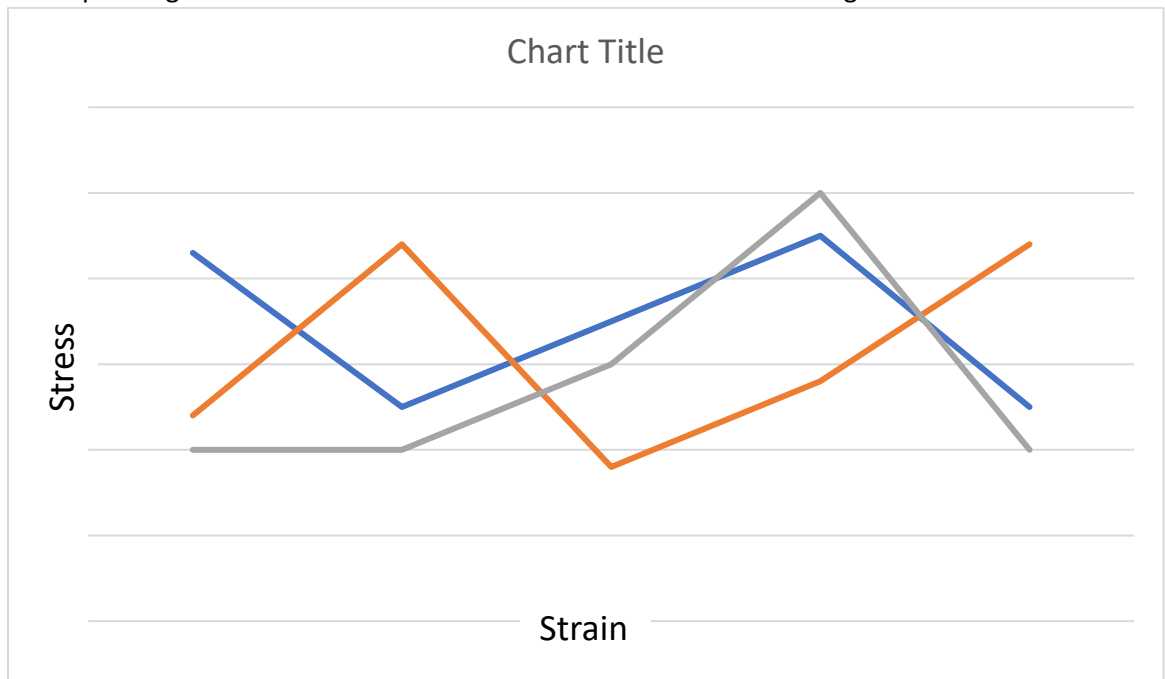
- (a) All groups' data for tensile test experiment will be provided on Moodle. Using the provided data files, plot engineering stress-strain curves in the same figure. Comment on possible reasons behind observed differences. (You can use Excel, Matlab, Python, or anything else you want to plot)



- (b) Using the data, calculate mean values for: Young's Modulus (E), 0.2% offset yield strength (σ_0), and ultimate tensile strength (σ_u). Compare with the values provided in the experimental report files.

(c) Comment on the dependence of the yield strength on the amount of offset. In particular, compare the yield strength values for offset values of 0.1%, 0.2% and 0.3%.

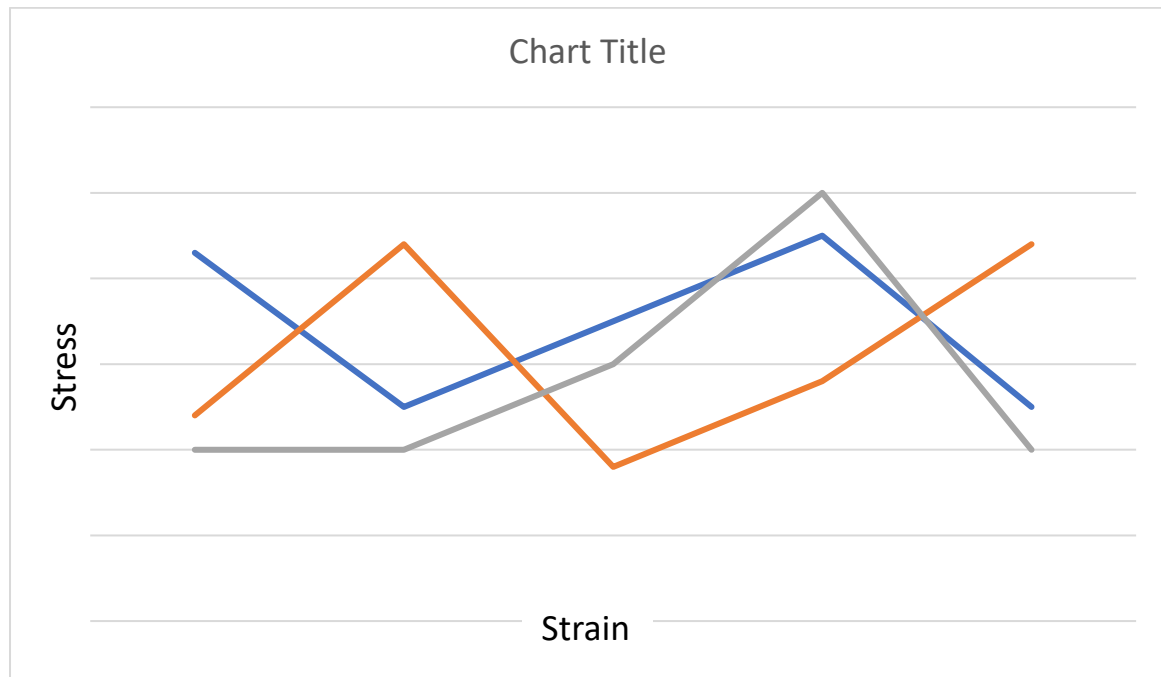
(d) Choose one of the engineering stress-strain curves from part (a) and calculate the corresponding true stress-strain curve. Plot both curves on the same figure.



(e) Search online about Lüders bands and explain why this concept is relevant to the test results.

Specimen 2: Polymer

- (a) All groups' data for tensile test experiment is provided on Moodle. Using the provided data files, plot engineering stress-strain curves in the same figure.



- (b) Using the provided data, calculate a mean value for ultimate tensile strength (σ_u). Compare with the values provided in the experimental report files.

- (c) Give a physics-based explanation of the observed differences between the stress-strain curves for the metal and polymer specimens.

(d) Comment on the qualitative and quantitative dependence of the polymer curve on the strain-rate.

(e) How does the necking behavior of polymer differ from the necking behavior observed for steel?